

PROJECT DOCUMENTATION

STREAMLINING IT PROCUREMENT:

AUTOMATING STANDARD LAPTOP ORDERS WITH FLOW DESIGNER

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Project Name:

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow
Designer

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1. INTRODUCTION

1.1 Project Overview

IT procurement is an important process in an organization because employees require suitable IT equipment to perform their daily work. One of the common requirements is providing standard laptops to employees. The procurement process generally involves request submission, approval, task assignment, procurement processing, and final fulfilment.

When these activities are handled manually, the process can become time-consuming. Employees may have to wait for approvals, contact different teams for updates, and repeatedly follow up on their requests. IT and procurement teams may also have to perform repetitive activities such as creating tasks, assigning tasks, updating statuses, and communicating with employees.

To overcome these challenges, this project proposes an automated standard laptop procurement workflow using the ServiceNow platform and Flow Designer.

The project titled “Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer” focuses on automating the workflow for a standard laptop request.

The employee submits a laptop request through ServiceNow. Once the request is submitted, Flow Designer automatically controls the required workflow. The process includes approval, task creation, task assignment, procurement processing, status updates, notifications, and request completion.

The automation reduces unnecessary manual intervention and provides a structured method for processing laptop requests. It also improves request visibility and helps the responsible teams manage procurement activities efficiently.

Main Workflow

Employee	Laptop Request	Flow Designer	Approval	IT/Procurement Task	Task
	Assignment	Laptop Fulfilment	Status Update	Completion	

The project demonstrates the practical use of ServiceNow Flow Designer for automating an IT procurement process.

1.2 Purpose

The main purpose of this project is to automate the standard laptop procurement process using ServiceNow Flow Designer.

The system is designed to reduce manual processing and provide a structured workflow from the initial request to final completion.

The major purposes of the project are:

1. To simplify the standard laptop request process.
2. To automate the approval workflow.
3. To automatically create required procurement or IT tasks.
4. To assign tasks to the appropriate team or user.
5. To reduce repetitive manual activities.
6. To provide better visibility of request status.
7. To improve communication through notifications.
8. To maintain request information in a centralized ServiceNow environment.
9. To provide a consistent and organized procurement workflow.

2. IDEATION PHASE

2.1 Problem Statement

In an organization, employees may require laptops for their work. The process of obtaining a laptop can involve several stages, including submitting a request, obtaining approval, assigning the request to the responsible team, processing the order, and completing the request.

When these activities are performed manually, several problems can occur. Request processing may take more time, approvals may be delayed, and tasks may need to be manually created and assigned. Employees may also have limited visibility of the progress of their requests.

IT and procurement teams may need to repeatedly perform similar activities for every laptop request.

Manual communication can also result in delays and information gaps.

Therefore, there is a need for an automated system that can manage the standard laptop procurement process in a structured and consistent manner.

The proposed project uses ServiceNow Flow Designer to automate the workflow.

The proposed process is:

Employee Request Approval Automated Task Creation Task Assignment Procurement
 Processing Status Update Completion

2.2 Empathy Map Canvas

User Persona

Primary User: Employee requesting a standard laptop

The employee needs a standard laptop for work and expects the procurement process to be simple, fast, transparent, and easy to track.

Think and Feel

- "I need a laptop for my work."
- "I want the request process to be simple."
- "I do not want unnecessary delays."
- "I want to know the current status of my request."
- "I should not have to repeatedly contact the IT team."

See

The employee may see different stages such as request submission, approval, processing, and fulfilment.

Hear

The employee may receive information from the manager, IT team, procurement team, or system notifications regarding the request.

Say and Do

The employee submits the request, provides the required information, waits for approval, checks the status, and receives the laptop after the request is processed.

Pain Points

- Manual request processing
- Approval delays
- Repeated follow-up
- Limited status visibility
- Manual task assignment
- Communication gaps
- Repetitive activities

Needs and Gains

- Faster processing
- Automated workflow execution
- Status visibility
- Reduced manual effort
- Better communication

Key Insight

Employees need a simple, transparent, and automated laptop procurement process that reduces unnecessary waiting and follow-up.

2.3 Brainstorming

Brainstorming was performed to identify possible solutions for improving the standard laptop procurement process.

Several ideas were considered:

1. Online laptop request submission
2. Automated approval workflow
3. Flow Designer automation
4. Automatic task creation

5. Automatic task assignment

6. Automated status updates

7. Notification system

8. Centralized request tracking

9. Request history maintenance

10. Automated request completion

The selected idea was to automate standard laptop orders using ServiceNow Flow Designer.

The selected solution provides a structured workflow connecting request submission, approval, task creation, assignment, processing, status updates, and completion.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The Customer Journey Map describes the different stages experienced by an employee while requesting a standard laptop.

1. Identifying the Need

The employee identifies the need for a standard laptop to perform work-related activities.

2. Submitting the Request

The employee submits the laptop request through ServiceNow and provides the required information.

3. Request Review and Approval

The submitted request is processed through the defined approval workflow. The required approver reviews the request.

4. Procurement Processing

After approval, the automated workflow creates the required IT or procurement task. The responsible team processes the laptop order.

5. Laptop Fulfilment

The laptop procurement process is completed and the laptop is prepared for the employee.

6. Request Completion

The request status is updated after the procurement process is completed. The employee receives the required status update.

Customer Pain Points

- Manual processing
- Approval delays
- Repeated follow-up
- Limited visibility
- Manual task assignment
- Communication gaps

Opportunities for Improvement

- Automated request processing
- Automated approval
- Automatic task creation
- Automatic task assignment
- Status tracking
- Notifications

3.2 Solution Requirement

Functional Requirements

The system should:

1. Allow employees to submit a standard laptop request.
2. Capture required employee and laptop request information.
3. Create a request record in ServiceNow.
4. Trigger the Flow Designer workflow.
5. Process the required approval.
6. Automatically create procurement or IT tasks.
7. Assign tasks to the responsible team or user.
8. Update the request status.
9. Generate required notifications.
10. Maintain request information for tracking.

Non-Functional Requirements

Usability: The request process should be simple and easy to understand.

Reliability: The workflow should execute consistently.

Security: Access to requests and related information should be controlled using appropriate permissions.

Efficiency: Automation should reduce unnecessary manual processing.

Maintainability: The workflow should be easy to modify when requirements change.

User Requirements

- Employee
- Approver
- IT/Procurement Team
- System Administrator

System Requirements

- ServiceNow instance
- Service Catalog
- Flow Designer
- Request and task records
- Approval configuration
- Notification configuration
- Appropriate roles and permissions
- Computer or laptop
- Internet connection
- Web browser

3.3 Data Flow Diagram

The Data Flow Diagram represents how information moves through the automated standard laptop procurement system.

Main Entities

- Employee
- ServiceNow
- Flow Designer
- Approver
- IT/Procurement Team

Main Data Flow

Employee Laptop Request ServiceNow Flow Designer Approval IT/Procurement Task
Status Update Employee

Level 0 DFD

Employee Standard Laptop Procurement System IT/Procurement Team

Level 1 DFD

1. Submit Laptop Request
2. Store Request Information
3. Trigger Flow Designer
4. Process Approval
5. Create Procurement Task
6. Assign Task
7. Process Laptop Order
8. Update Request Status
9. Notify Employee
10. Complete Request

Data Stored

- Employee details
- Laptop request details
- Approval information

- Task information
- Request status
- Processing details

3.4 Technology Stack

ServiceNow

ServiceNow is the primary platform used for implementing the project. It provides the environment required to create and manage the laptop procurement workflow.

Service Catalog

Service Catalog provides the interface through which employees can submit the standard laptop request.

Flow Designer

Flow Designer is the main automation component of the project. It controls the workflow after the laptop request is submitted.

It supports automated processes involving triggers, actions, conditions, approvals, tasks, notifications, and record operations.

ServiceNow Tables

ServiceNow tables store and manage request, task, user, approval, and status-related information.

Notifications

Notifications are used to communicate important events such as request submission, approval, processing, and completion.

Roles and Permissions

Roles and permissions help control access to the required records and functions.

Overall Technology Flow

ServiceNow Service Catalog Laptop Request Flow Designer Approval IT/Procurement Task
Status Update Notification

4. PROJECT DESIGN

4.1 Problem Solution Fit

The project was designed to address the major problems identified in the manual laptop procurement process.

The existing process may involve manual request handling, approval delays, repetitive task assignment, communication gaps, and difficulty tracking request progress.

The proposed ServiceNow solution addresses these problems through workflow automation.

Flow Designer automatically controls the required activities after a request is submitted. Approval can be processed through the workflow, and required tasks can be created and assigned automatically.

The solution also provides better tracking because request and task information are maintained within ServiceNow.

Expected Benefits

- Reduced manual processing
- Better request tracking
- Improved task management
- Reduced repetitive work
- Better communication
- Consistent workflow
- Improved process efficiency

4.2 Proposed Solution

The proposed solution uses ServiceNow and Flow Designer to automate the standard laptop procurement process.

Working of the Proposed Solution

1. Request Submission

The employee submits a standard laptop request through ServiceNow.

2. Request Creation

ServiceNow creates and stores the request information.

3. Flow Trigger

Flow Designer is triggered when the required condition is satisfied.

4. Approval

The request is sent to the appropriate approver.

5. Task Creation

After the required approval, the system automatically creates the required IT or procurement task.

6. Task Assignment

The task is assigned to the appropriate responsible team or user.

7. Procurement Processing

The IT or procurement team processes the laptop order.

8. Status Update

The request status is updated according to the progress of the process.

9. Notification

The employee receives relevant notifications regarding the request.

10. Completion

After the procurement process is completed, the request is marked as completed.

4.3 Solution Architecture

Employee Layer

The employee initiates the process by submitting a standard laptop request.

ServiceNow Request Layer

ServiceNow receives the request and stores the required information.

Flow Designer Layer

Flow Designer controls the automated workflow and executes the required actions.

Approval Layer

The approval stage ensures that the request is reviewed before further processing.

IT/Procurement Layer

The responsible team receives the automatically generated task and processes the laptop order.

Data Layer

ServiceNow records store request, task, approval, user, and status information.

Notification Layer

Notifications communicate important workflow updates to the employee.

Overall Architecture

Employee	ServiceNow	Service Catalog	Laptop Request	Flow Designer	Approval	IT/
Procurement Task		Laptop Fulfilment	Status Update	Notification	Completion	

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Project planning defines the activities required to successfully develop and complete the automated standard laptop procurement project.

Project Objective

The main objective is to design and implement an automated standard laptop procurement workflow using ServiceNow Flow Designer.

Project Planning Activities

Requirement Analysis

The requirements of the project are identified and analyzed.

Problem Identification

The problems associated with manual laptop procurement are studied.

Solution Design

The proposed automated workflow and architecture are designed.

ServiceNow Configuration

The required ServiceNow components are configured.

Flow Designer Development

The automation flow is created and configured.

Testing

The request submission, approval, task creation, task assignment, status updates, and notifications are tested.

Error Correction

Errors identified during testing are analyzed and corrected.

Final Demonstration

The complete automated laptop procurement workflow is demonstrated.

Documentation

The final project documentation is prepared with diagrams, testing details, results, advantages, disadvantages, conclusion, and future scope.

Project Workflow

Requirement Analysis Problem Identification Solution Design ServiceNow Configuration
Flow Development Testing Error Correction Demonstration Documentation

Resources Required

- ServiceNow instance
- Service Catalog
- Flow Designer
- Computer or laptop
- Internet connection
- Web browser

- Documentation tools

Risk Planning

Possible risks include:

- Incorrect workflow configuration
- Approval failures
- Task assignment problems
- Notification failures
- Testing errors

These risks can be reduced through proper configuration, repeated testing, validation, and correction of identified issues.

Expected Project Outcome

The expected outcome is a working ServiceNow workflow that automates standard laptop procurement and reduces unnecessary manual processing.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Performance Testing is performed to verify whether the automated laptop procurement workflow operates efficiently and reliably.

The testing focuses on the ServiceNow request process and Flow Designer workflow.

Testing Objectives

1. To verify system response during request submission.
2. To verify Flow Designer execution.
3. To check approval processing.

4. To verify automatic task creation.
5. To verify task assignment.
6. To check status updates.
7. To verify notifications.
8. To identify workflow delays or errors.
9. To verify consistent system performance.

Testing Process

The testing begins by submitting a standard laptop request through ServiceNow.

The request creation and system response are observed. After submission, the Flow Designer workflow is monitored to ensure that it starts correctly.

The approval stage is tested, followed by task creation and assignment. The request status is checked during each stage.

Notifications are also verified to ensure that important updates are communicated correctly.

The complete workflow is then tested from request submission to completion.

Functional Verification

The following functions are verified:

- Laptop request submission
- Request record creation
- Flow Designer trigger
- Approval
- Task creation
- Task assignment

- Status update
- Notification
- Request completion

Performance Factors

The main factors considered are:

- System response time
- Flow execution
- Approval processing
- Task creation
- Task assignment
- Status update
- Notification delivery
- Workflow reliability

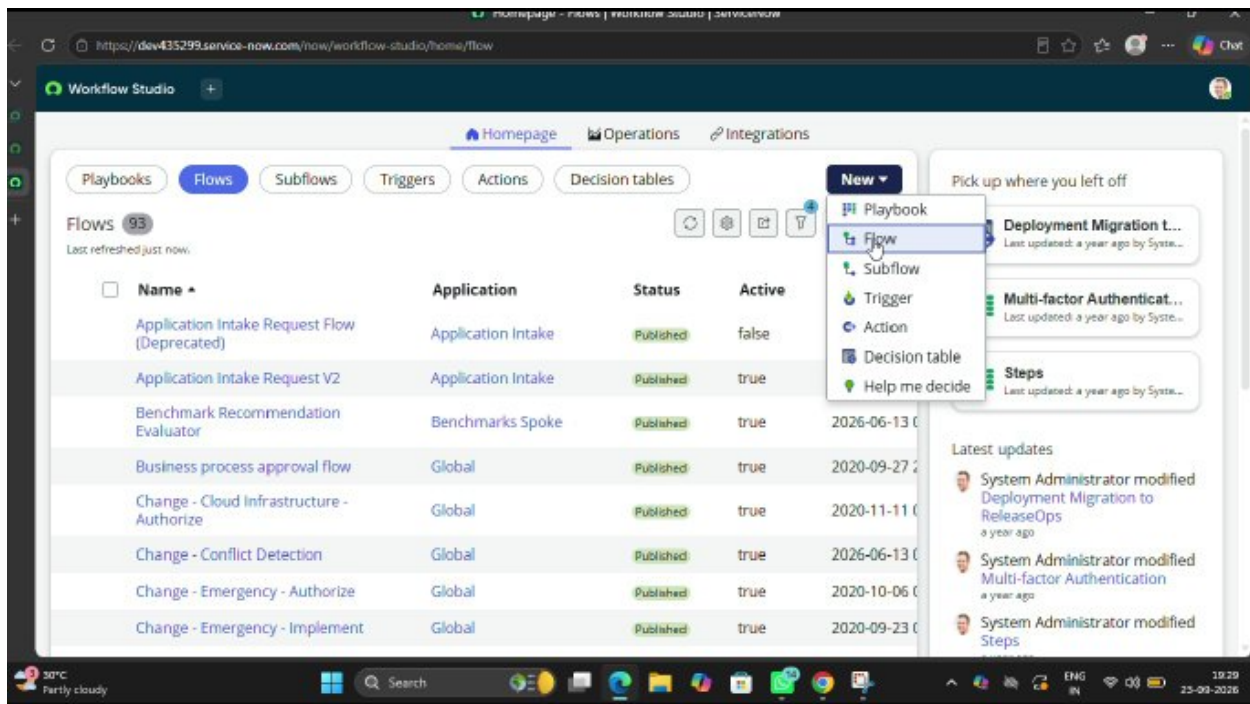
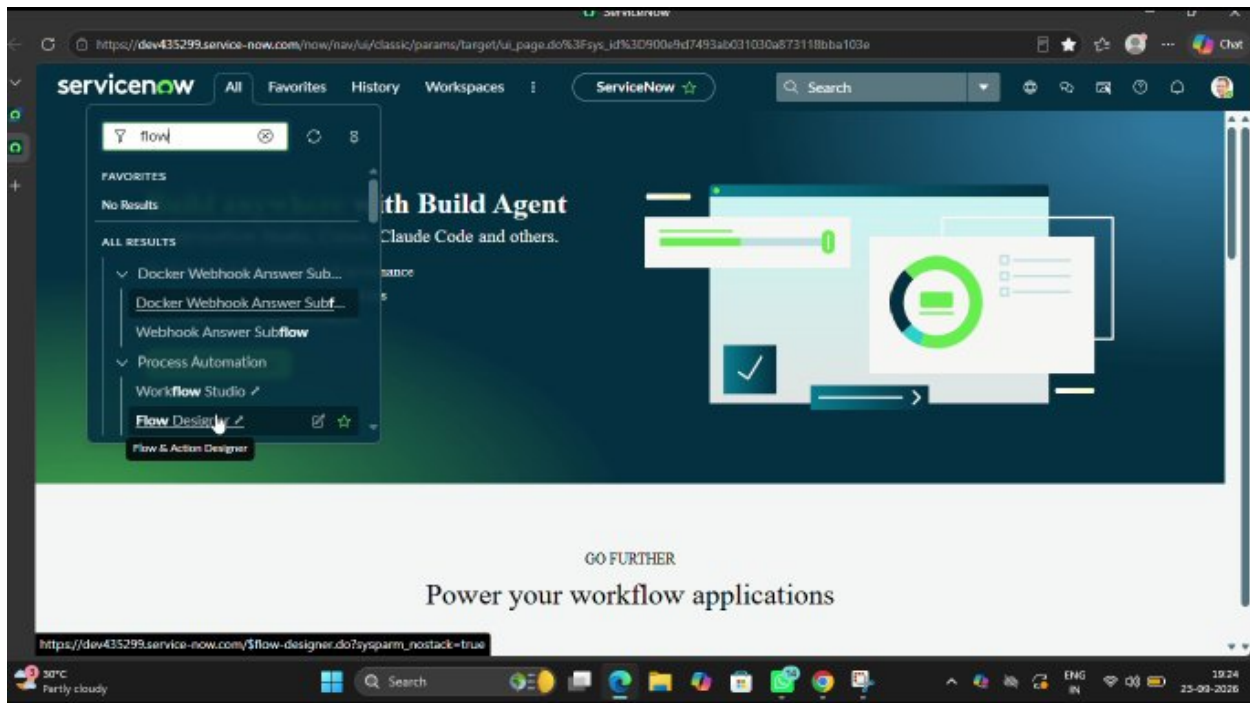
Expected Outcome

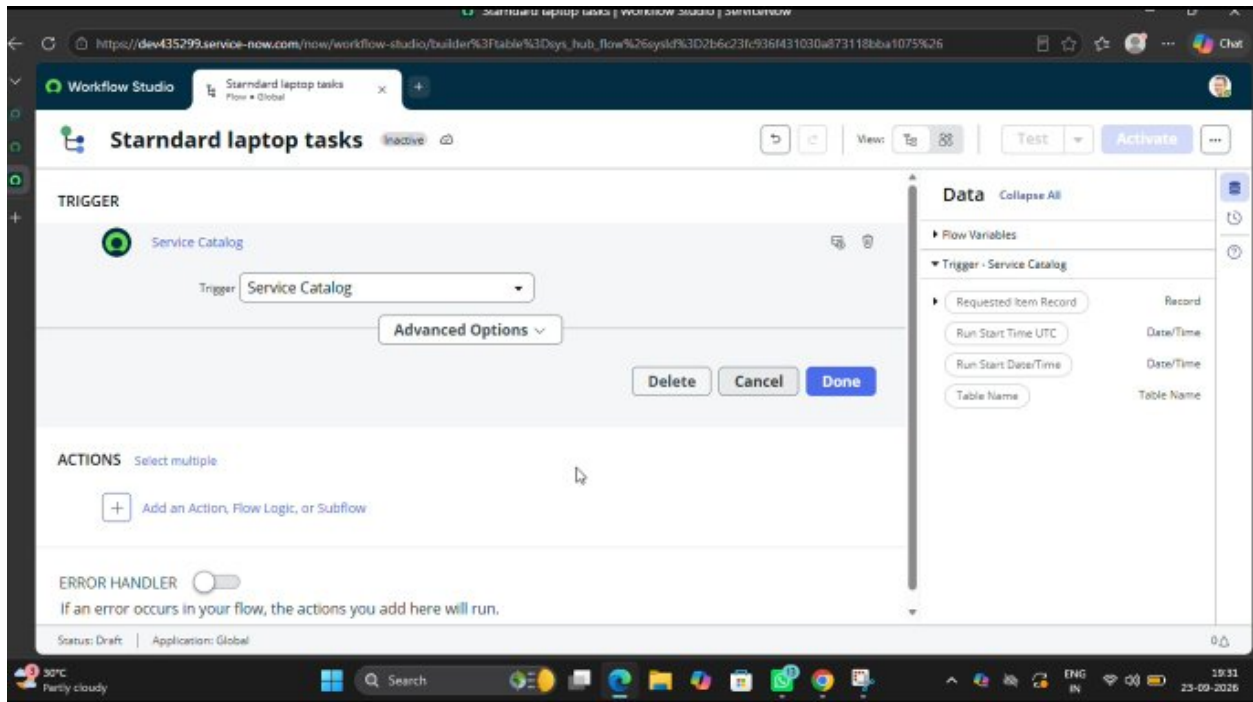
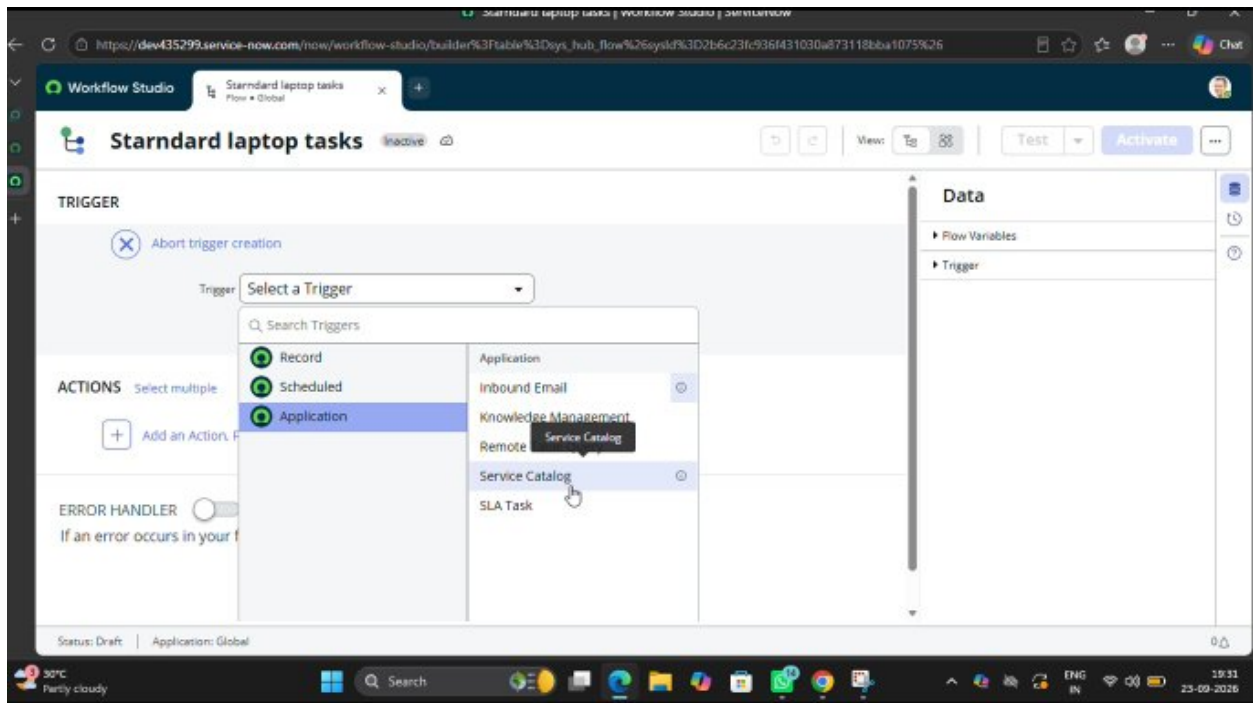
The system should process laptop requests smoothly and execute the required automated workflow without unnecessary delays or failures.

7. RESULTS

7.1 Output Screenshots

The following section contains screenshots of the implemented project in the ServiceNow environment. The screenshots demonstrate the standard laptop request process, Flow Designer automation, approval, task processing, status updates, and completion of the request.





Workflow Studio | Starndard laptop tasks | Flow: Global

Starndard laptop tasks

TRIGGER: Service Catalog

ACTIONS: Select multiple

ERROR HANDLING: If an error occurs...

Search: crea

- Most Recent
 - VA Telemetry Daily Rollup U... Global
 - VA Telemetry Daily Rollup U... Global
 - Look Up Records ServiceNow Core
 - Check Attachment Readiness Global
 - Run script include on FDH... Global
- Popular
- INSTALLED SPOKES
 - ServiceNow Core
 - AES Catalog Builder
 - AES Flow Templates
 - AES Mobile Templates
 - APS Portal Template

Data Panel:

- Flow Variables
- Trigger - Service Catalog
 - Requested Item Record (Record)
 - Run Start Time UTC (Date/Time)
 - Run Start Date/Time (Date/Time)
 - Table Name (Table Name)

Status: Draft | Application: Global

Workflow Studio | Starndard laptop tasks | Flow: Global

Starndard laptop tasks

Action Properties

Action: Create Catalog Task

Action Inputs

Table Name: Catalog Task [sc_task]

* Requested Item (Requested Item): Trigger ... Requested Item ...

Short Description: Laptop need to Configured

Fields:

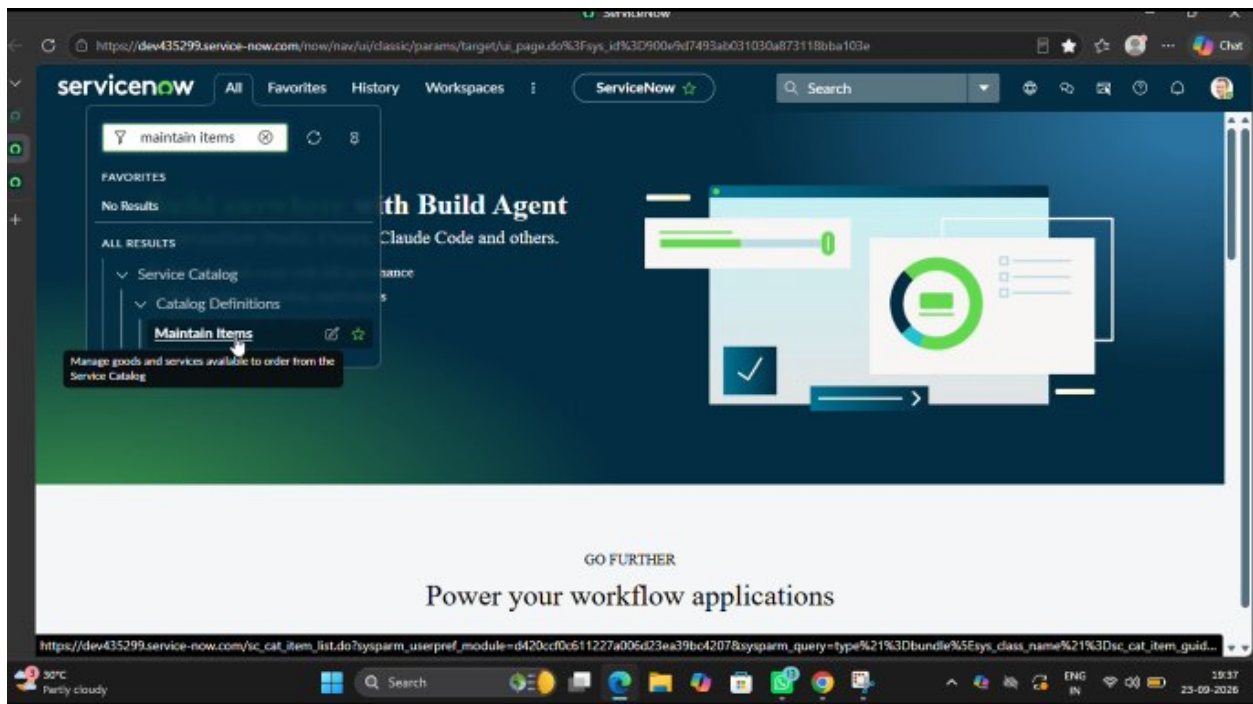
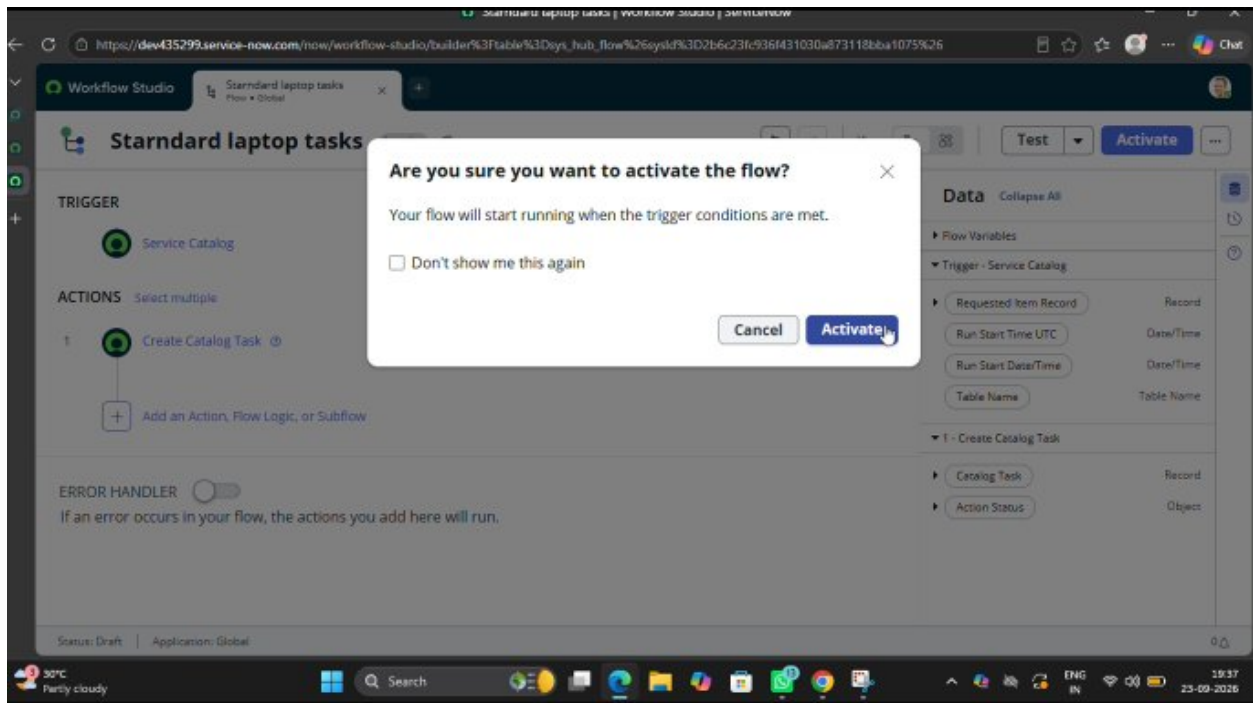
- Description: Laptop need to Configured
- Assignment group: Hardware
- Approval: Approved

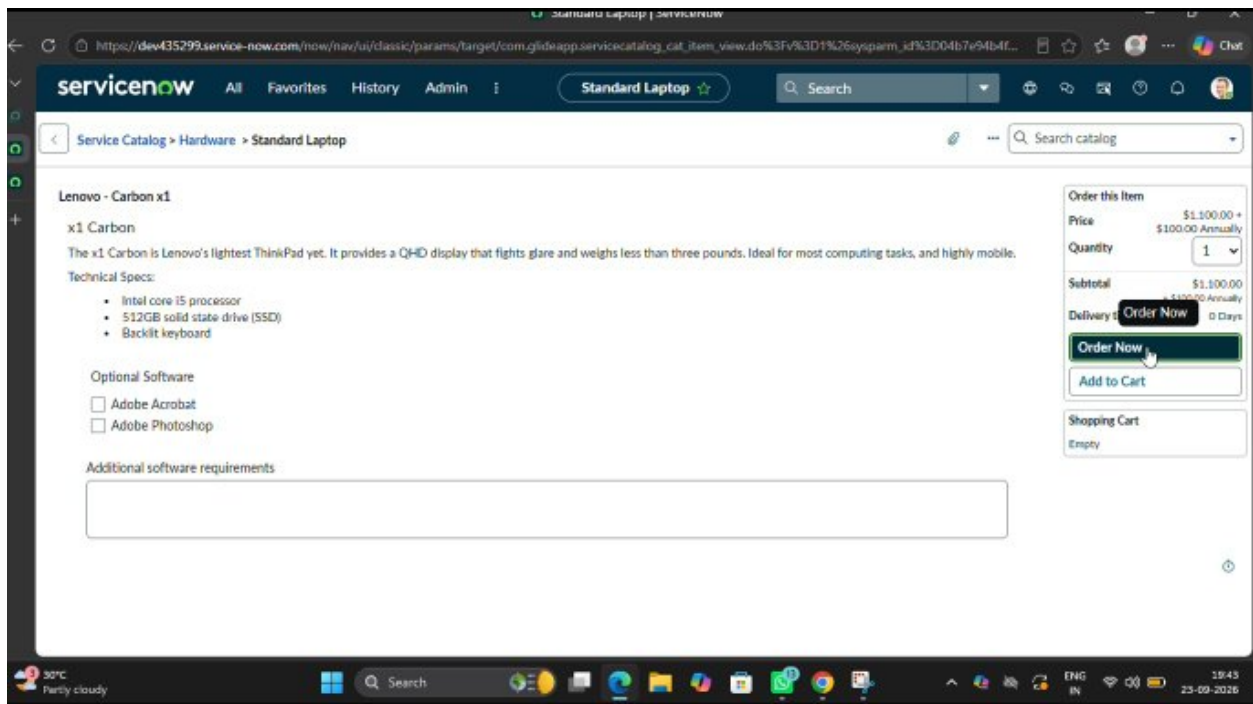
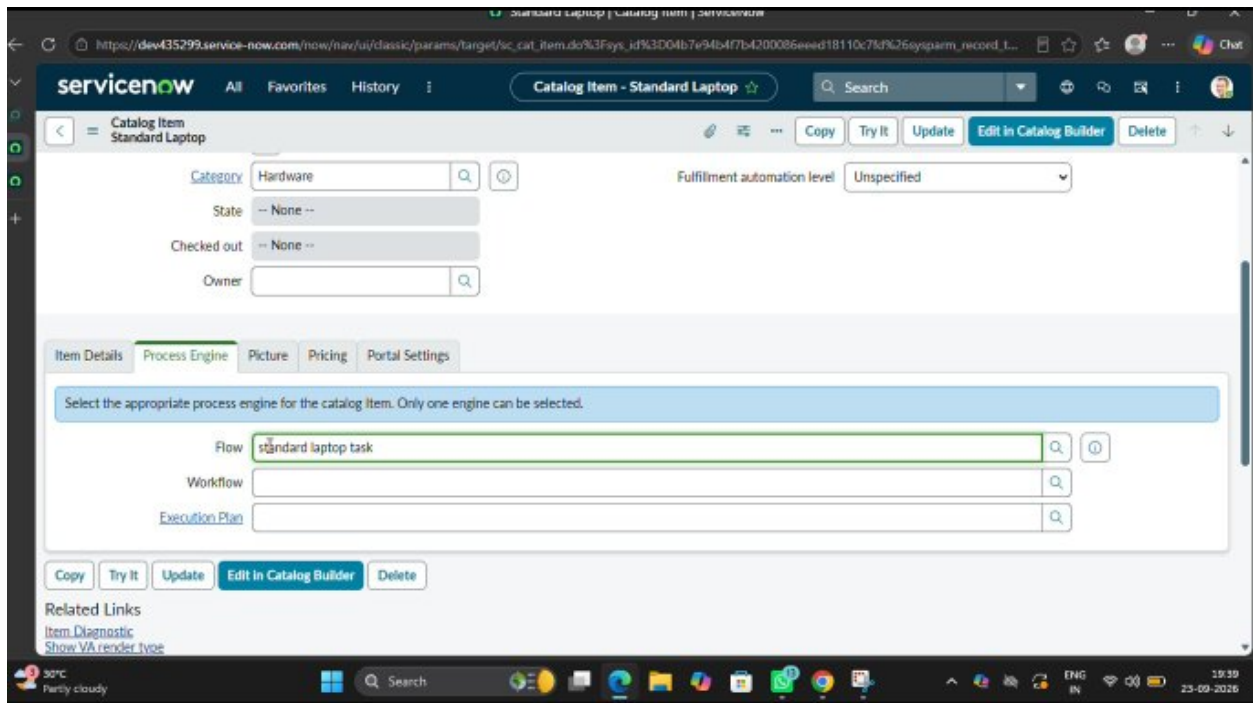
+ Add field value

Data Panel:

- Flow Variables
- Trigger - Service Catalog
 - Requested Item Record (Record)
 - Run Start Time UTC (Date/Time)
 - Run Start Date/Time (Date/Time)
 - Table Name (Table Name)
- 1 - Create Catalog Task
 - Catalog Task (Record)
 - Action Status (Object)

Status: Modified | Application: Global





servicenow | All | Favorites | History | Admin | Order Status: REQ0010001 | Search

Order Status

Thank you, your request has been submitted

Order Placed: 2026-09-23 07:13:25
Request Number: REQ0010001 ☆
Estimated Delivery Date of Complete Order: 2026-09-23

Description (Includes Annual Charges)	Delivery Date	Stage	Price (ea.)	Quantity	Total
lenovo - Carbon x1	2026-09-23		\$1,100.00 +\$100.00 Annually	1	\$1,100.00 +\$100.00 Annually
Total					\$1,100.00 +\$100.00 Annually

Back to Catalog Continue Shopping Home

https://dev435299.servicenow.com/sc_request.do?sys_id=B5623b493af431030a873118bba1096&sysparm_record_target=sc_request&sysparm_catalog=e0d08b13c3330100c8b837659ba8fb4&sysparm_ca...

30°C Partly cloudy Search 19:43 23-09-2026

servicenow | All | Favorites | History | Admin | Request - REQ0010001 | Search

Request - REQ0010001 | Discuss | Follow | Update | Add New Item | Cancel Request | Copy | Delete

Activities: 1

System Administrator

Field changes: 2026-09-23 07:13:23

Impact: 3 - Low
Opened by: System Administrator
Priority: 4 - Low
State: Open

Update Add New Item Cancel Request Copy Delete

Requested Items (1) Approvers (1) Recurring Prices (1)

State Search Actions on selected rows... Edit

Approval for = REQ0010001

State	Approver	Comments	Created
Request	Eric Schroeder		2026-09-23 07:13:32

1 to 1 of 1

30°C Partly cloudy Search 19:44 23-09-2026

Request - REQ0010001

Activities: 1

System Administrator

Impact: 3 - Low
Opened by: System Administrator
Priority: 4 - Low
State: Open

Field changes: 2026-09-23 07:13:23

Update Add New Item Cancel Request Copy Delete

Requested Items (1)

State

Approval for = REQ0010001

State

Requested

Approver

Comments

Created

Eric Schroeder

2026-09-23 07:13:32

1 to 1 of 1

Requested Item - RITM0010001

Impact: 3 - Low
Opened by: System Administrator
Priority: 4 - Low
State: Open

Update Delete

Related Links

Flow Context

Catalog Tasks (1) Approvers Group approvals

Number Search

Request item = RITM0010001

Number	Assignment group	Assigned to	Short description	Actual start	Actual end	Order
SCTASK0010001	Hardware	(empty)	Laptop need to Configured	2026-09-23 07:15:09	(empty)	

1 to 1 of 1

The screenshot shows a ServiceNow Catalog Task form for SCTASK0010001. The form is titled 'Catalog Task - SCTASK0010001' and has buttons for Discuss, Follow, Update, Close Task, and Delete. The form includes the following fields:

- Number: SCTASK0010001
- Assigned to: (empty)
- Configuration item: (empty)
- Active: ☒
- Approval: Approved
- Priority: 4 - Low
- State: Open
- Request item: RJTM0010001
- Requested for: System Administrator
- Short description: Laptop need to Configured
- Description: Laptop need to Configured
- Work notes: Work notes

The 'Activities' section shows one activity by System Administrator with an impact of Low. The form is titled 'Catalog Task - SCTASK0010001' and has buttons for Discuss, Follow, Update, Close Task, and Delete.

8. ADVANTAGES & DISADVANTAGES

Advantages

Reduced Manual Work

The automation reduces repetitive manual activities for IT and procurement teams.

Faster Processing

Automated workflow activities reduce unnecessary delays between different stages of the procurement process.

Automatic Task Creation

Required tasks can be created automatically after the defined conditions are satisfied.

Better Task Management

Tasks can be assigned to the appropriate team or user, making responsibilities clearer.

Improved Tracking

The request status can be monitored within ServiceNow.

Better Communication

Automated notifications help communicate important updates to employees and responsible teams.

Centralized Information

Request, approval, task, and status information can be maintained within the ServiceNow environment.

Consistent Workflow

A predefined workflow can be followed for standard laptop requests, providing a consistent procurement process.

Disadvantages

Platform Dependency

The solution depends on the availability and configuration of the ServiceNow platform.

Initial Configuration Effort

Creating and configuring the workflow requires initial development and testing effort.

Limited to Standard Requests

The project is mainly designed for standard laptop procurement and may require additional configuration for non-standard equipment.

Configuration Errors

Incorrect Flow Designer conditions or actions can affect workflow execution.

Maintenance Requirement

The workflow may require updates when organizational procurement policies or requirements change.

9. CONCLUSION

The project "Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer" demonstrates how workflow automation can be used to improve the standard laptop procurement process.

The project addresses common challenges associated with manual procurement, including repetitive work, approval delays, manual task assignment, communication gaps, and limited request visibility.

By using ServiceNow and Flow Designer, the project provides a structured workflow for handling standard laptop requests. The employee can submit a request, after which the system can manage

approval, task creation, task assignment, status updates, notifications, and completion.

The project also demonstrates the importance of requirement analysis, solution design, planning, testing, and documentation when developing an automated business process.

The implemented solution provides a systematic approach to standard laptop procurement and demonstrates the practical use of ServiceNow Flow Designer for IT process automation.

10. FUTURE SCOPE

The project can be further enhanced with additional automation and integration features.

Automated Inventory Checking

The system can be enhanced to check laptop inventory automatically before processing a request.

Vendor Integration

The system can be integrated with external vendors or procurement systems to automate purchase-related activities.

Advanced Approval Rules

Different approval workflows can be introduced based on employee department, designation, laptop type, cost, or business requirements.

Multiple Device Categories

The system can be expanded to support other IT equipment such as monitors, keyboards, docking stations, tablets, and other standard devices.

Automated Notifications

More detailed notifications can be added for different stages such as approval, procurement, shipping, delivery, and completion.

Dashboard and Reporting

A dashboard can be introduced to display the number of requests, pending approvals, completed orders, processing time, and other procurement information.

AI-Based Automation

Future versions can use AI-based features to classify requests and assist with procurement-related activities.

Integration with Asset Management

The procurement workflow can be integrated with asset management so that the issued laptop can automatically be recorded as an organizational asset.

These enhancements can make the system more comprehensive and suitable for larger organizational IT procurement processes.

11. APPENDIX

The appendix provides additional supporting information and project resources.

Appendix A – ServiceNow Configuration

This section contains the configuration details used for implementing the automated standard laptop procurement process in ServiceNow.

Appendix B – Flow Designer Workflow

This section provides information about the Flow Designer workflow used to automate the standard laptop procurement process.

The workflow includes request triggering, approval, task creation, task assignment, status updates, notifications, and request completion.

Appendix C – Project Demonstration Video

The project demonstration video shows the working of the automated standard laptop procurement workflow, including request submission, approval, Flow Designer execution, task processing, status updates, and completion.

Video Link:

<https://drive.google.com/file/d/1Mpgs5oGhULAgOVxDhS7cDKq-gQeAwOub/view?usp=sharing>

Appendix D – GitHub Repository

The GitHub repository contains the project-related files and supporting resources, if applicable.

GitHub Link:

https://github.com/razhiyaraihana61-gif/Streamlining-IT-procurement_NM-Project.git

Appendix E – Additional Project Details

This section contains additional project-related information, requirements, implementation details,

and supporting material used during the development of the project.
